

Heat Treatment of Plain-Carbon Steels

Effects of carbon content, quench medium, and tempering temperature on hardness

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Course	ME 215-102, NJIT	Instructor	Prof. Suwattananont
Team	Annemarie Malek (lead), Luis Ramirez, Samuel Kitoye-Rufus, Hayley Hernandez, Abdullah Masood, Stacy Aquino	Tools	Box furnaces, water/oil/air quench, Rockwell hardness tester, forging press

PROJECT SUMMARY

This project investigated how heat treatment changes the mechanical properties of low- and high-carbon steels. Samples were heated to 200 °C, 500 °C, and 960 °C, quenched in water, oil, or air, and in a second week tempered at 200, 400, and 500 °C, with Rockwell C hardness measured before and after each step. Heating to 960 °C and water quenching formed martensite and produced the largest hardness gains: about 3× for low-carbon steel (9.9 to 29.5 HRC) and about 3.2× for high-carbon steel (19.1 to 61.0 HRC). Water quenching produced the hardest structure and air cooling the softest, while higher tempering temperatures progressively traded hardness for toughness. An aluminum billet was also softened and forged to demonstrate metal forming.

1. Introduction & Objectives

Steel’s strength, durability, and adaptability make it one of the most important engineering materials. This project examined how heat treatment — heating, quenching, and tempering — changes the structure and mechanical properties of low- and high-carbon steels. The objectives were to measure how temperature and cooling rate affect hardness, strength, and ductility; to study martensite formation and its dependence on carbon content; and to evaluate how well different hardening treatments improve strength and durability.

2. Background

Heat treatment is the controlled heating and cooling of metals to alter their internal structure and properties without changing their shape. It is essential in production because it can soften a material for easier forming or harden it for strength and wear resistance. **Annealing** softens the metal and makes it easier to work; **quenching** cools it rapidly to form hard martensite; and **tempering** restores toughness and reduces brittleness after quenching (Callister & Rethwisch, 2018).

The iron–carbon phase diagram shows which phases are stable at each temperature and composition, while the time–temperature–transformation (TTT) diagram shows how cooling rate controls which phases form. Together they allow engineers to predict and control final properties and select the heat treatment that gives the right balance of strength, hardness, and ductility for an application.

3. Procedure

Week 1 — heating and quenching. After reviewing the iron–carbon diagram and how quench media affect microstructure, the team set up the Rockwell tester, selected the C scale, and measured the initial hardness of three low-carbon and seven high-carbon samples (three readings each). Low-carbon samples were heated to 200, 500, and 960 °C for 30 minutes and water quenched. High-carbon samples were heated to the same temperatures; those at 960 °C were quenched in water, oil, or air. Samples were lightly sanded and re-tested.

Week 2 — tempering and forming. Three water-quenched high-carbon samples were tempered for 30 minutes at 200, 400, and 500 °C, cooled to room temperature, and re-tested. An aluminum billet was also heated to 400 °C for 30 minutes and forged in a press. All data sheets were reviewed and signed off by the TA.

4. Results

Table 1. Average Rockwell C hardness before and after heating and quenching (3 readings per sample)

Sample	Treatment	Initial HRC (avg)	After treatment HRC (avg)	Change
Low C 1	200 °C, water	10.2	9.5	−0.7
Low C 2	500 °C, water	11.8	11.8	0.0
Low C 3	960 °C, water	9.9	29.5	+19.6
High C 1	200 °C, water	15.5	19.7	+4.2
High C 2	500 °C, water	17.2	16.7	−0.5
High C 3–5	960 °C, water	14.6–19.2	58.3–61.0	+40 to +44
High C 6	960 °C, oil	19.3	57.8	+38.5
High C 7	960 °C, air	18.9	44.5	+25.6

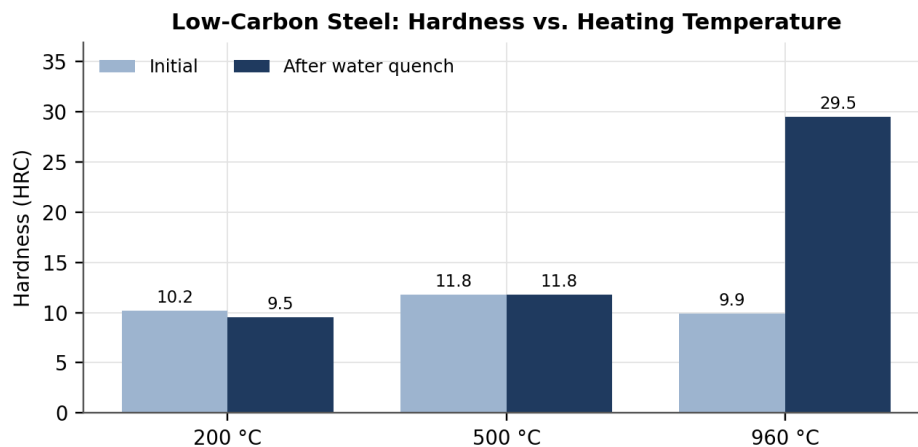


Figure 1. Low-carbon steel hardness before and after heating and water quenching.

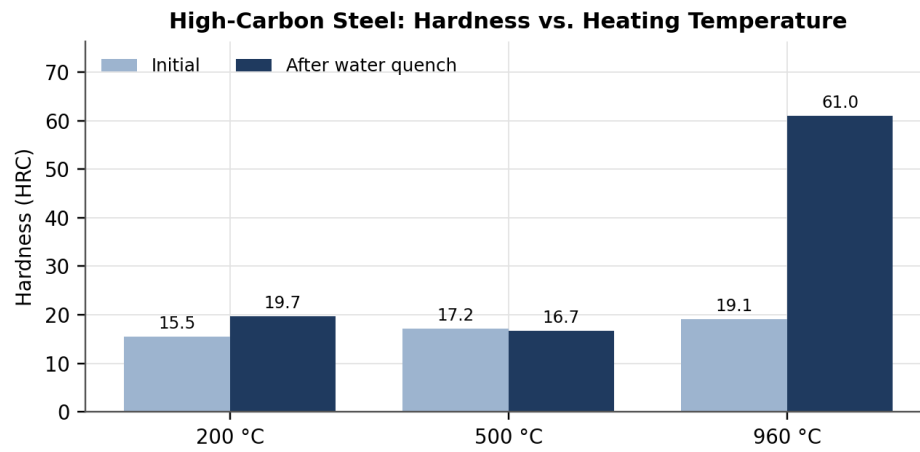


Figure 2. High-carbon steel hardness before and after heating and water quenching.

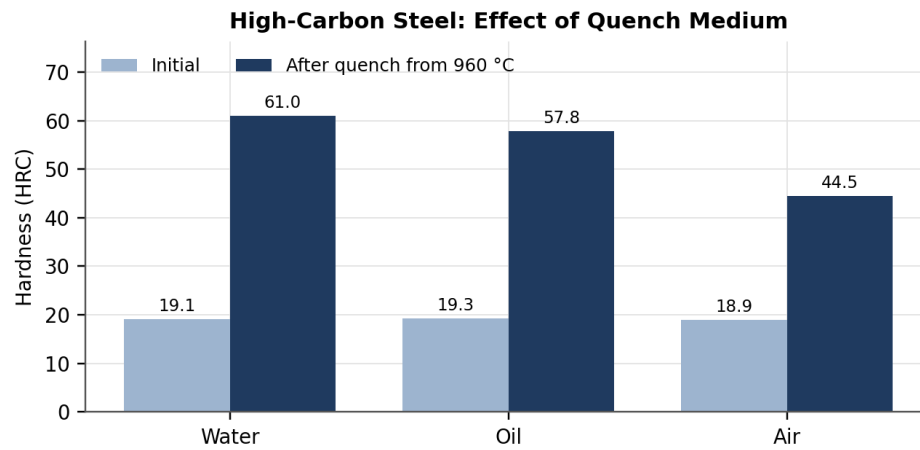


Figure 3. Effect of quench medium on high-carbon steel quenched from 960 °C.

Table 2. Tempering of water-quenched high-carbon steel (30 min)

Tempering temperature	As-quenched HRC	After tempering HRC	Change
200 °C	58.3	57.2	-1.1
400 °C	61.0	47.2	-13.8
500 °C	60.0	39.0	-21.0

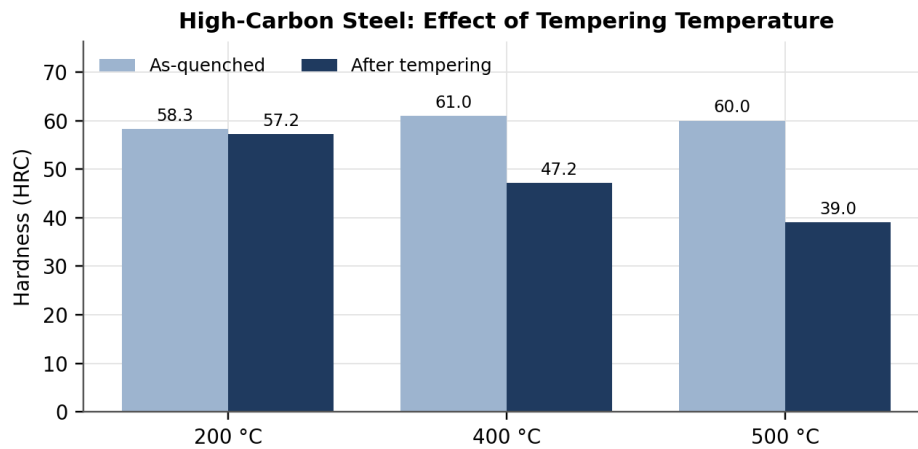


Figure 4. Hardness of quenched high-carbon steel before and after tempering.

Table 3. Tensile strength estimated from hardness

Condition	Approx. HRC	Est. tensile strength (MPa)
Low-carbon, initial	10.5	440
Low-carbon, 960 °C water quench	29.5	≈ 960
High-carbon, initial	18	690
High-carbon, 960 °C water quench	61	2,205

The aluminum billet was also formed in the press: the unheated billet measured 0.435 in in diameter after pressing (from 0.5 in), while the billet heated to 400 °C was reduced further, to 0.380 in, showing how heating lowers the force needed to deform the metal.

5. Discussion

Carbon content. Before treatment, the high-carbon samples were consistently harder than the low-carbon samples (roughly 17 vs. 10 HRC on average). Carbon atoms distort the iron lattice and restrict dislocation motion, so hardness rises with carbon content. High-carbon steel therefore suits strength- and wear-critical applications, while low-carbon steel is better for machining and forming because of its lower hardness and greater ductility.

Heating temperature. At 200 °C and 500 °C, both steels changed only slightly, since neither temperature is high enough to form austenite; the small shifts are consistent with minor atomic rearrangement, stress relief, or early recovery. At 960 °C, both steels transformed to austenite, and water quenching converted it to martensite, producing a dramatic jump in hardness. The effect was far greater in the high-carbon steel (61 HRC vs. 29.5 HRC), because more dissolved carbon produces a harder, more strained martensite.

Quench medium. Cooling rate controlled final hardness. Water, the most severe quench, produced the hardest structure (about 61 HRC). Oil cooled more slowly and more uniformly, giving slightly lower hardness (57.8 HRC) with less risk of distortion and cracking. Air cooling was slowest and produced the softest result (44.5 HRC), consistent with a mix of softer phases rather than full martensite. Variation within the oil-quenched readings suggests that cooling uniformity and small differences in composition also play a role.

Tempering. Tempering lowered hardness in proportion to temperature. At 200 °C, hardness barely changed (–1.1 HRC), indicating stress relief with little change to the martensite. At 400 °C, hardness dropped by about 14

HRC as martensite partially transformed to softer tempered structures, and at 500 °C the largest drop occurred (–21 HRC). This confirms the goal of tempering: higher temperatures increase ductility and toughness at the cost of hardness. During the forming exercise, the press initially did not operate correctly, but adjusting the handle and mechanism resolved the issue.

6. Conclusion

The results showed that carbon content and cooling method dominate the mechanical properties of steel. High-carbon steel was consistently harder than low-carbon steel, confirming the role of carbon in strength. Heating to 960 °C followed by a water quench produced the hardest structure, making the choice of quench medium critical, while tempering converted part of the martensite into tougher structures that reduced brittleness while retaining useful hardness. Overall, the project demonstrated how heat treatment can be tailored to refine the properties of carbon steel for real engineering and industrial applications.

7. Key Concepts

What are normalizing, full annealing, and process annealing?

Normalizing heats steel above its critical temperature and air cools it, refining grain structure and relieving internal stress to give a uniform structure with improved strength and toughness. **Full annealing** heats above the critical temperature, holds, and furnace cools; it lowers hardness and improves ductility and machinability while relieving stress. **Process annealing** heats cold-worked metal to below the critical temperature to restore ductility and relieve stress so it can be shaped further.

What is martensite?

A hard, brittle microstructure formed when steel is quenched rapidly from the austenite range. The fast cooling traps carbon atoms in the crystal lattice and distorts it, greatly increasing hardness but sharply reducing ductility.

Why do low- and high-carbon steels quenched from 960 °C reach different hardness?

High-carbon steel forms more, and harder, martensite because of its higher carbon content. Low-carbon steel forms less martensite and remains softer and more ductile after quenching.

Why does high-carbon steel quenched from 960 °C end up much harder than when quenched from 500 °C?

At 960 °C the steel is fully austenitic, and quenching converts that austenite to martensite. At 500 °C no austenite forms, so the original ferrite–pearlite structure remains and hardness barely changes.

Is cementite harder than pearlite?

Yes. Cementite (Fe_3C) is a hard, brittle iron carbide. Pearlite is a layered mix of soft ferrite and cementite, so its hardness falls well below that of cementite alone.

What is the purpose of tempering?

To reduce the brittleness of quenched steel and increase its toughness while retaining much of its strength and hardness.

Can Brinell and Rockwell values be compared?

Yes, by converting one to the other scale using standard conversion tables; the harder material will rate higher on either scale. Conversions are approximate, so they are best used for comparison rather than precise values.

Suggested heat treatment for a warped 1080 steel katana blade

(1) **Stress-relief anneal** below the critical temperature with slow cooling to relieve machining stresses and reduce warpage. (2) **Differential hardening**: heat and rapidly quench the edge to harden it. (3) **Temper** the blade to relieve quenching stresses and restore fracture toughness while keeping the edge hard.

What is case hardening?

A surface treatment that hardens the outer layer of a part while leaving the core softer and tougher.

References

Callister, W. D., Jr., & Rethwisch, D. G. (2018). *Materials science and engineering: An introduction* (10th ed.). Wiley.